FUNDAMENTALS OF PROGRAMMING

Lecture 13: STRUCTURE

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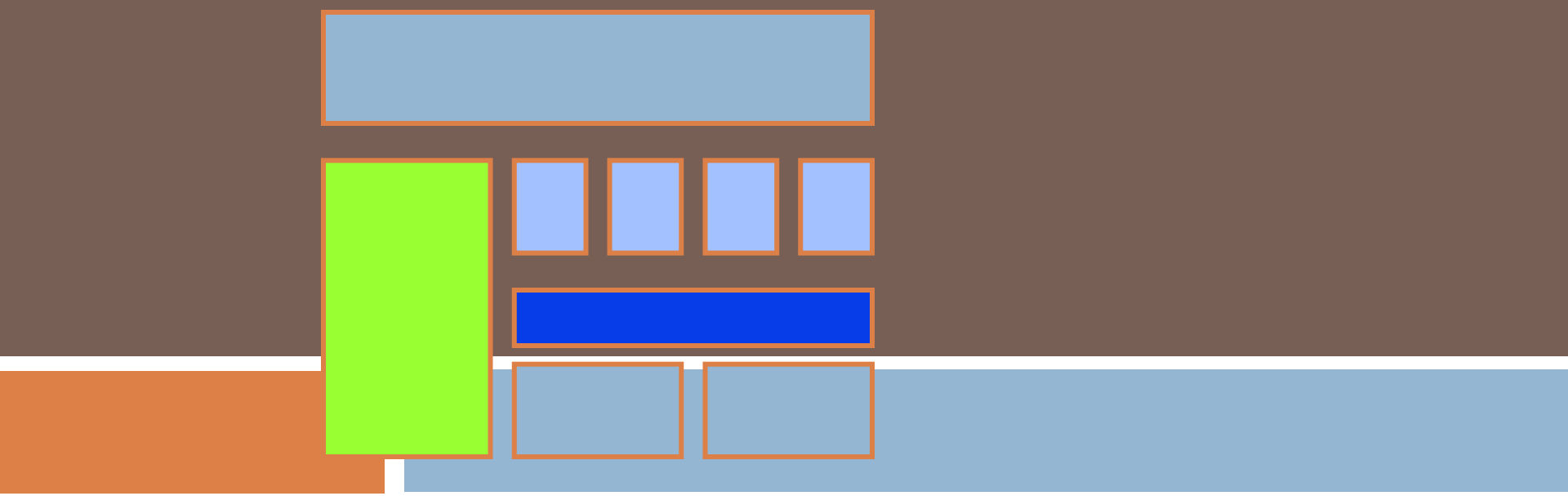
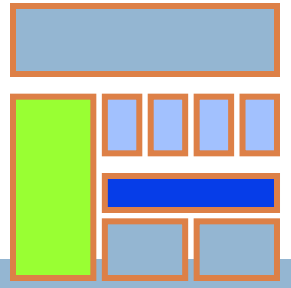
STRUCTURES

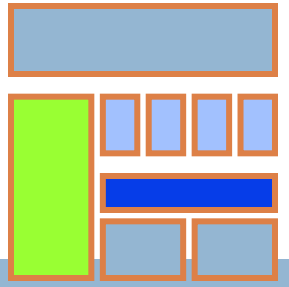


Structures



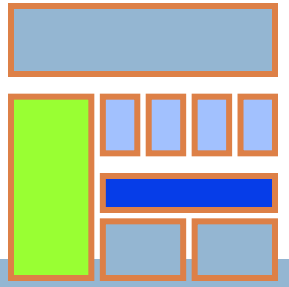
- A Structure is a collection of related data items, possibly of different types.
- A structure type in C++ is called struct.
- A struct is heterogeneous in that it can be composed of data of different types.
- In contrast, array is homogeneous since it can contain only data of the same type.

Structures



- Structures hold data that belong **together.**
- Examples:
  - Student record: student id, name, major, gender, start year, …
  - Bank account: account number, name, currency, balance, …
  - Address book: name, address, telephone number, …
- In database applications, structures are called records.

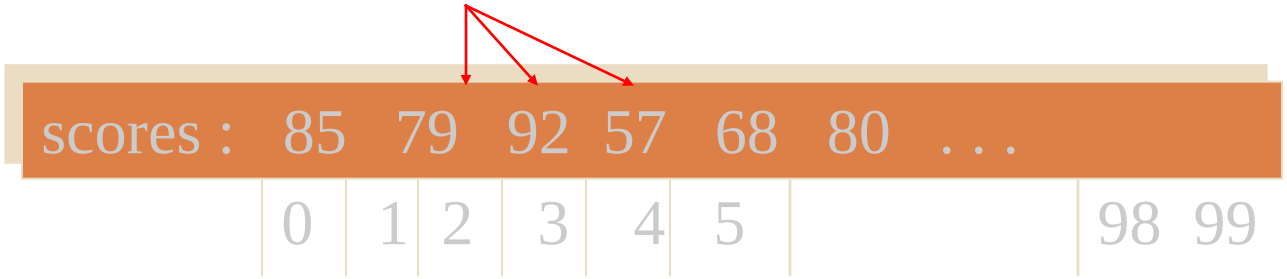
Structures



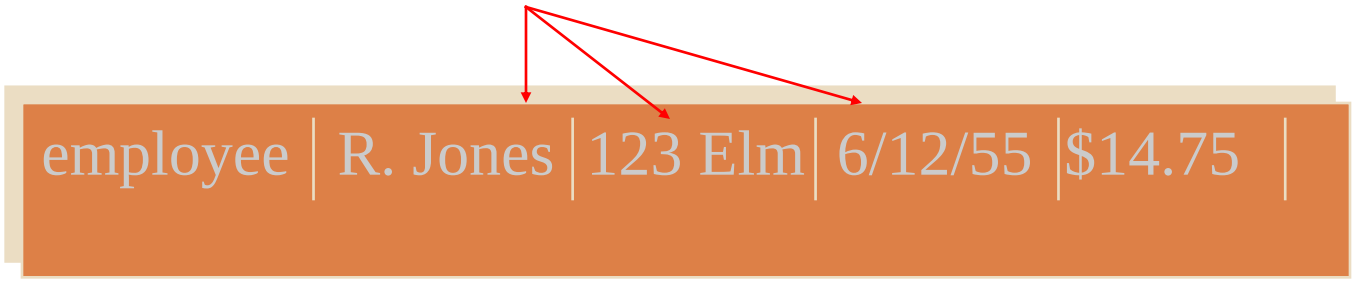
- Individual components of a struct type are called members (or fields).
- Members can be of different types (simple, array or struct).
- A struct is named as a whole while individual members are named using field identifiers.
- Complex data structures can be formed by defining arrays of structs.

Structures

- Recall that elements of arrays must all be of the same type



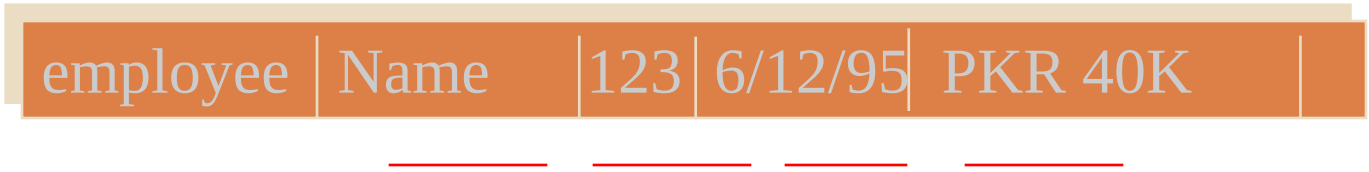
- In some situations, we wish to group elements of different types



Structures

- RECORDS are used to group related components of different types
- Components of the record are called fields

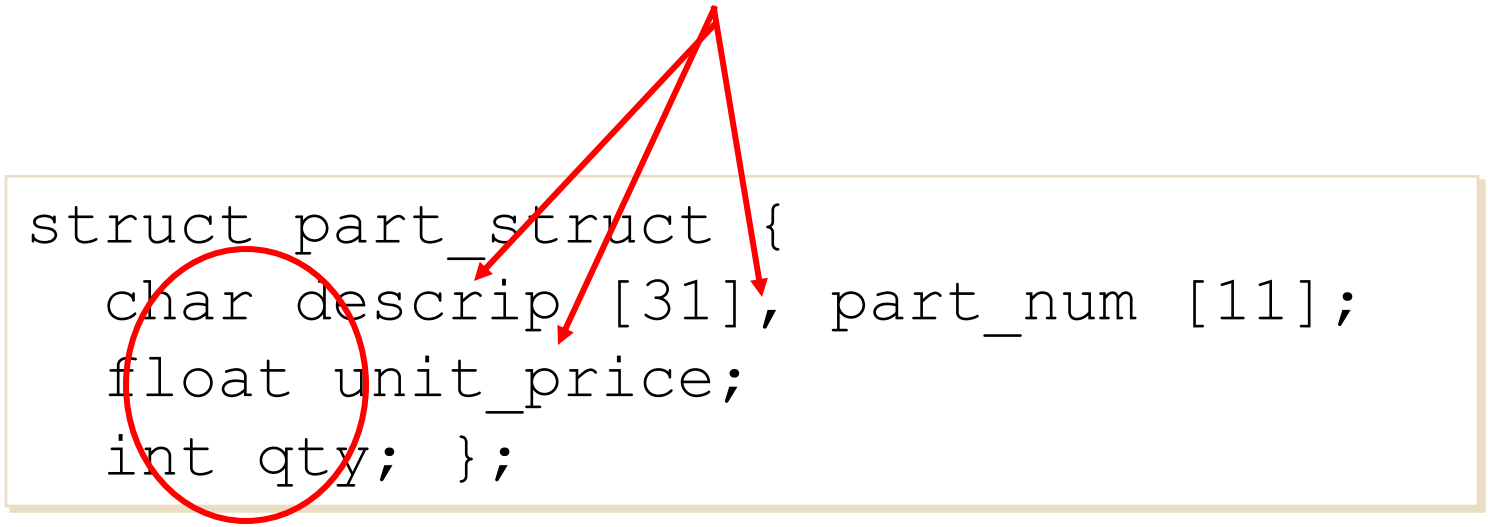
- In C++



  - record called a struct (structure)
  - fields called members

Structures

- C++ **struct**
  - structured data type
  - fixed number of components
  - elements accessed by name, not by index
  - components may be of different types



```
struct part_struct {
   char descrip [31], part_num [11];
   float unit_price;
   int qty; };
```

Declaring `struct` Variables

- Given

```
struct part_struct {
   char descrip [31], part_num [11];
   float unit_price;
   int qty; };
```

- Declare :

Use `struct` name as a type.

```
part_struct new_part, old_part;
```

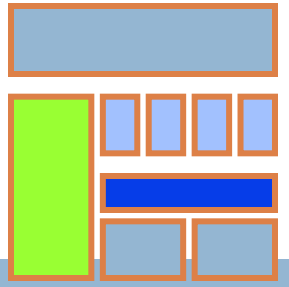
Accessing Components

- Use the name of the record
  the name of the member
  separated by a dot .

```
old_part.qty = 5;
cout << new_part.descrip;
```

- The dot is called the member selector

struct basics



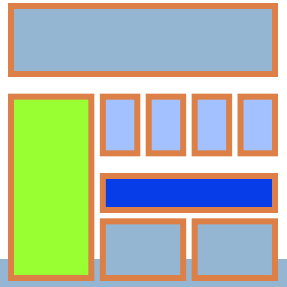
- Definition of a structure:

```
struct <struct-type>{
        <type> <identifier_list>;
        <type> <identifier_list>;
        ...
} ;
```

- Example:

```
struct Date {
        int day;
        int month;
        int year;
} ;
```

struct examples



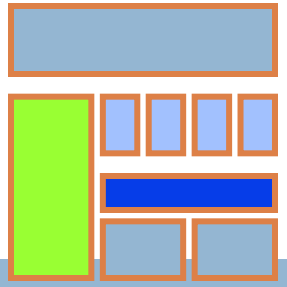
- Example:

```
struct StudentInfo{
    int Id;
    int age;
    char Gender;
    double CGA;
};
```

- Example:

```
struct StudentGrade{
    char Name[15];
    char Course[9];
    int Lab[5];
    int Homework[3];
    int Exam[2];
};
```

struct examples



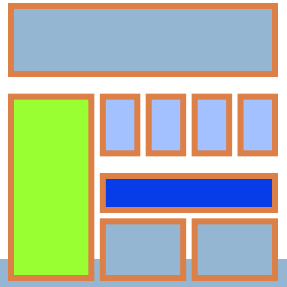
- Example:

```
struct BankAccount{
    char Name[15];
    int AcountNo[10];
    double balance;
    Date Birthday;
};
```

- Example:

```
struct StudentRecord{
    char Name[15];
    int Id;
    char Dept[5];
    char Gender;
};
```

struct basics



- Declaration of a variable of struct type:

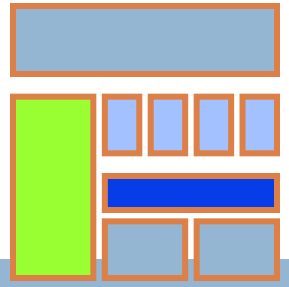
```
<struct-type> <identifier_list>;
```

- Example:

```
StudentRecord Student1, Student2;
```


Student1 and **Student2** are variables of **StudentRecord** type.

struct basics

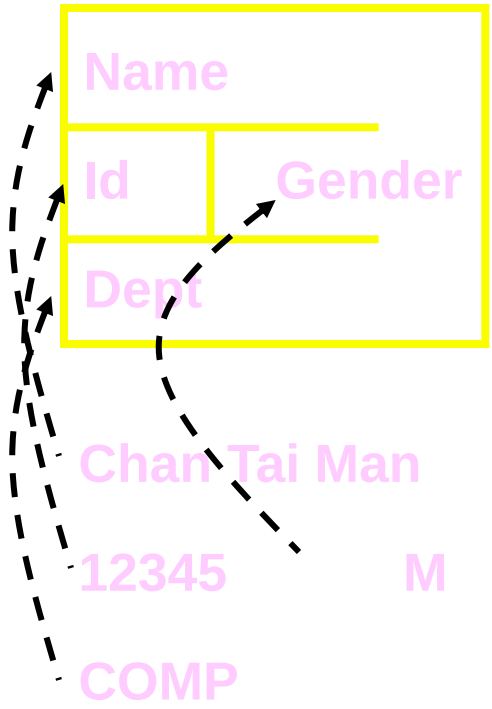


- The members of a struct type variable are accessed with the dot (.) operator:

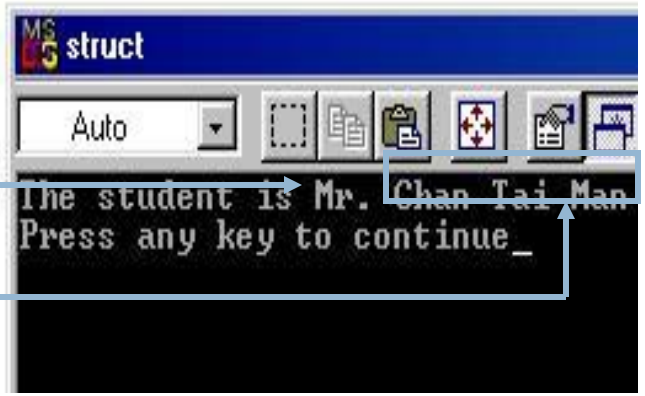
```
<struct-variable>.<member_name>;
```

- Example:

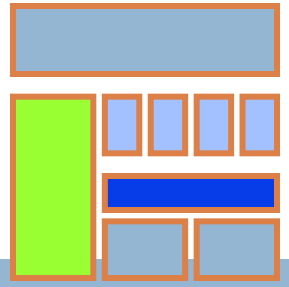
```
strcpy(Student1.Name, "Chan Tai Man");
Student1.Id = 12345;
strcpy(Student1.Dept, "COMP");
Student1.gender = 'M';
cout << "The student is ";
switch (Student1.gender){
    case 'F': cout << "Ms. "; break;
    case 'M': cout << "Mr. "; break;
}
cout << Student1.Name << endl;
```



```
#include <string.h>

struct StudentRecord {
        char Name[22];
        int Id;
        char Dept[22];
        char gender;
};

int main()  {
    StudentRecord Student1;

    strcpy(Student1.Name, "Chan Tai Man");
    Student1.Id = 12345;
    strcpy(Student1.Dept, "COMP");
    Student1.gender = 'M';

    cout << "The student is ";
    switch (Student1.gender){
        case 'F': cout << "Ms. "; break;
        case 'M': cout << "Mr. "; break;
    }
    cout << Student1.Name << endl;
    return 0;
}
```

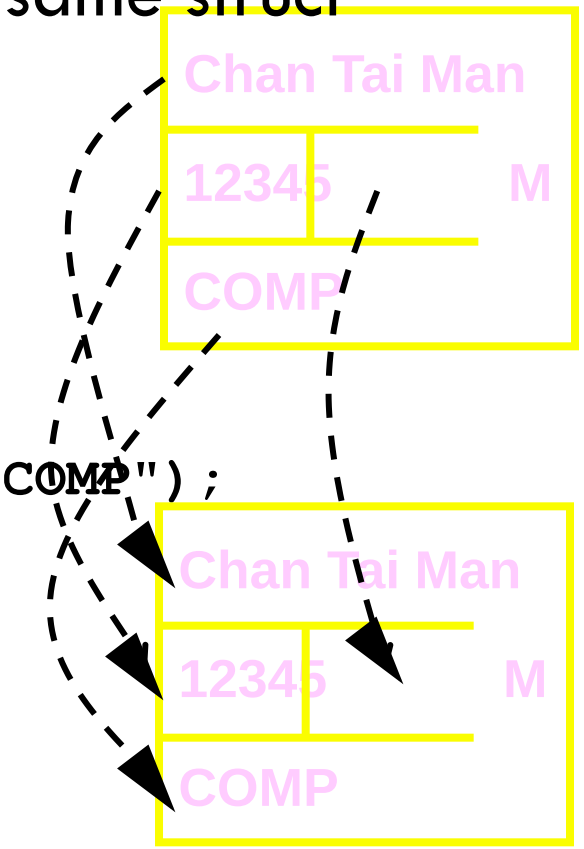


struct-to-struct assignment

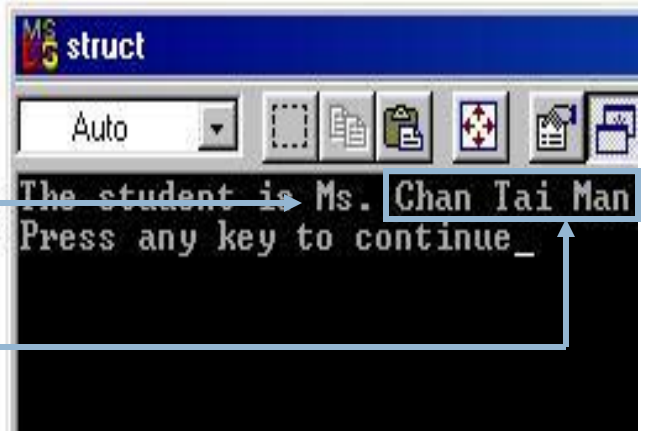


- The values contained in one struct type variable can be assigned to another variable of the same struct type.
- Example:

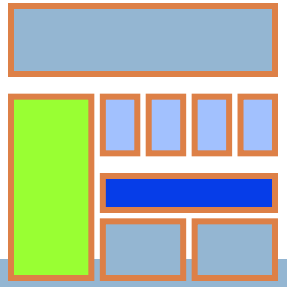
```
strcpy(Student1.Name,
       "Chan Tai Man");
Student1.Id = 12345;
strcpy(Student1.Dept, "COMP");
Student1.gender = 'M';

Student2 = Student1;
```



```
struct StudentRecord {
        char Name[22];
        int Id;
        char Dept[22];
        char gender;
};

int main()  {
    StudentRecord Student1, Student2;

    strcpy(Student1.Name, "Chan Tai Man");
    Student1.Id = 12345;
    strcpy(Student1.Dept, "COMP");
    Student1.gender = 'M';

    Student2 = Student1;
    Student2.gender = 'F';

    cout << "The student is ";
    switch (Student2.gender){
        case 'F': cout << "Ms. "; break;
        case 'M': cout << "Mr. "; break;
    }
    cout << Student2.Name << endl;
    return 0;
}
```

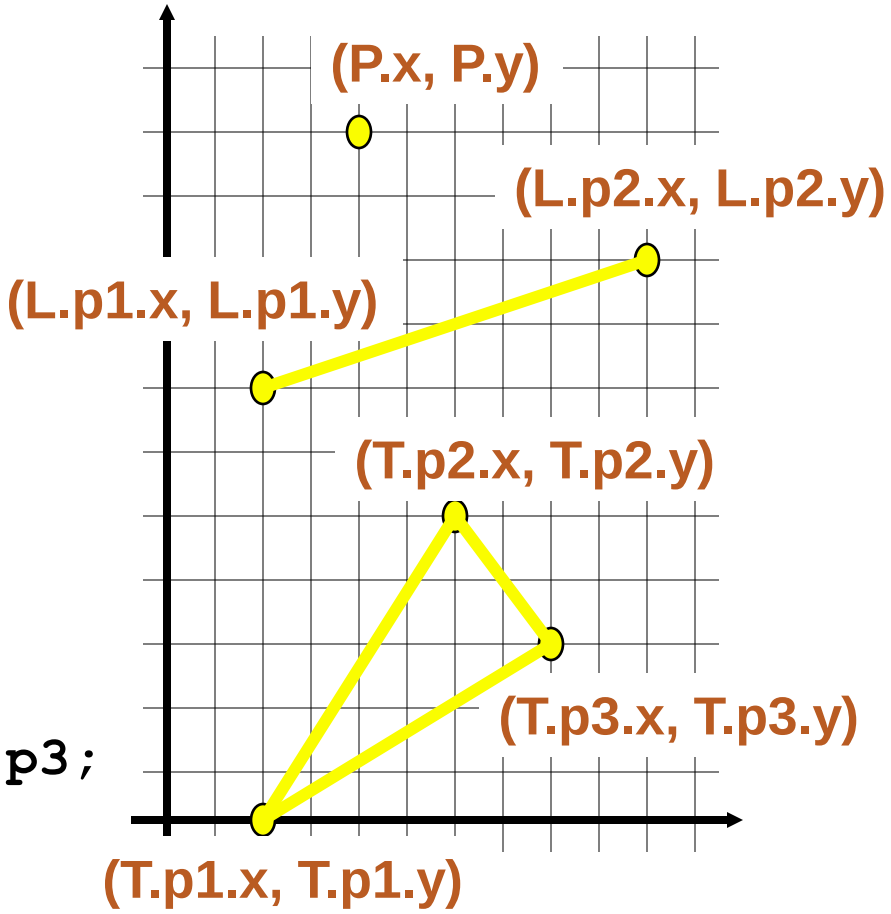


Nested structures

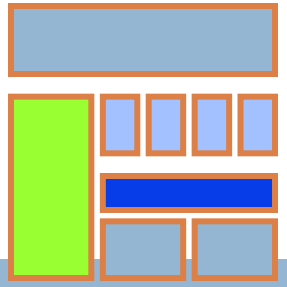


- We can nest structures inside structures.
- Examples:

```
struct point{
    double x, y;
};
point P;
struct line{
    point p1, p2;
};
line L;
struct triangle{
    point p1, p2, p3;
};
triangle T;
```

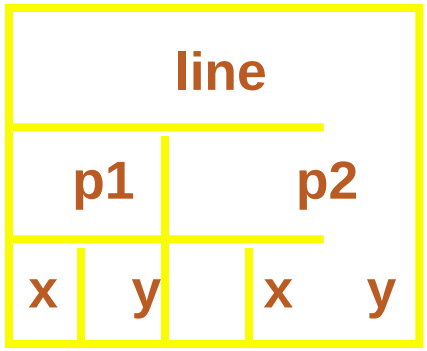


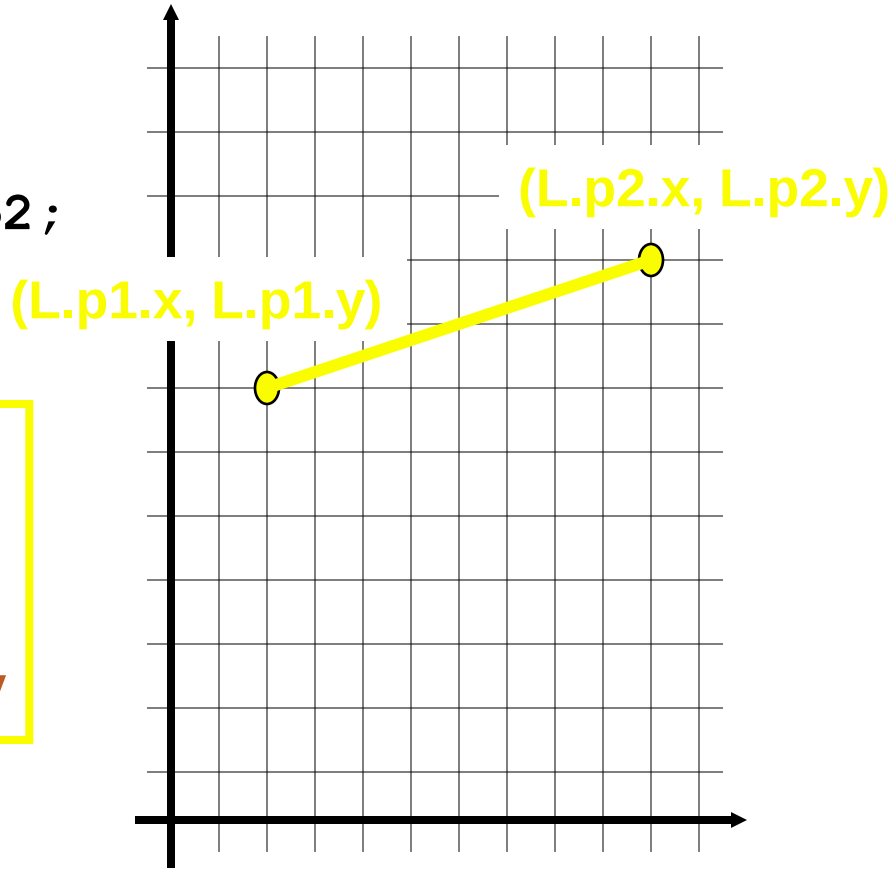
Nested structures



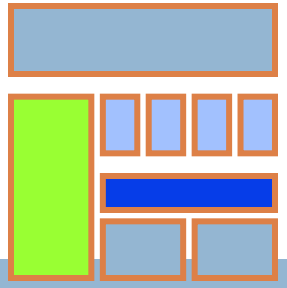
- We can nest structures inside structures.

```
struct line{
     point p1, p2;
};
line L;
```



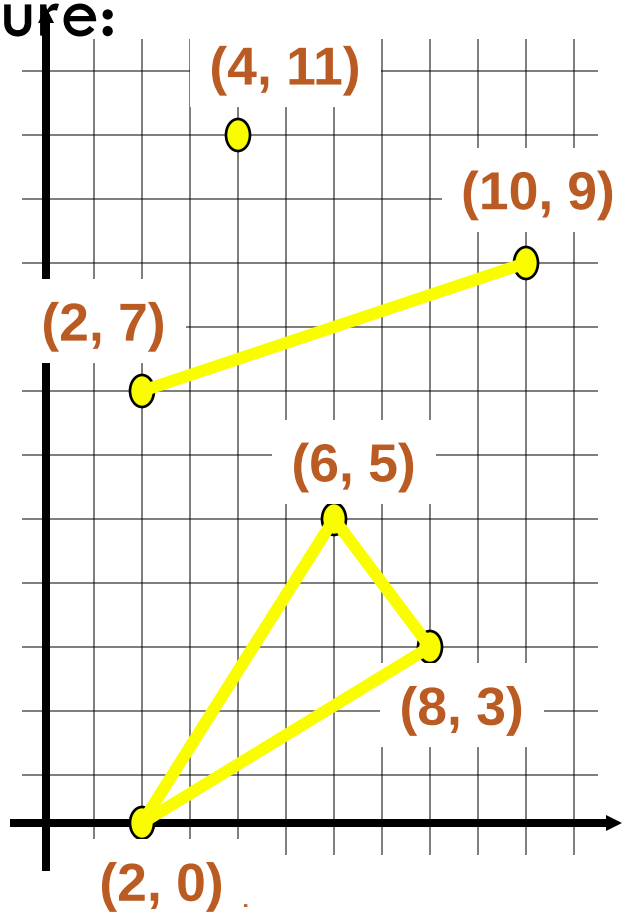


Nested structures

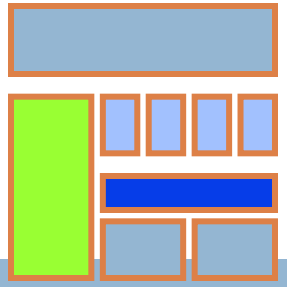


- Assign values to the variables `P`, `L`, and `T` using the picture:

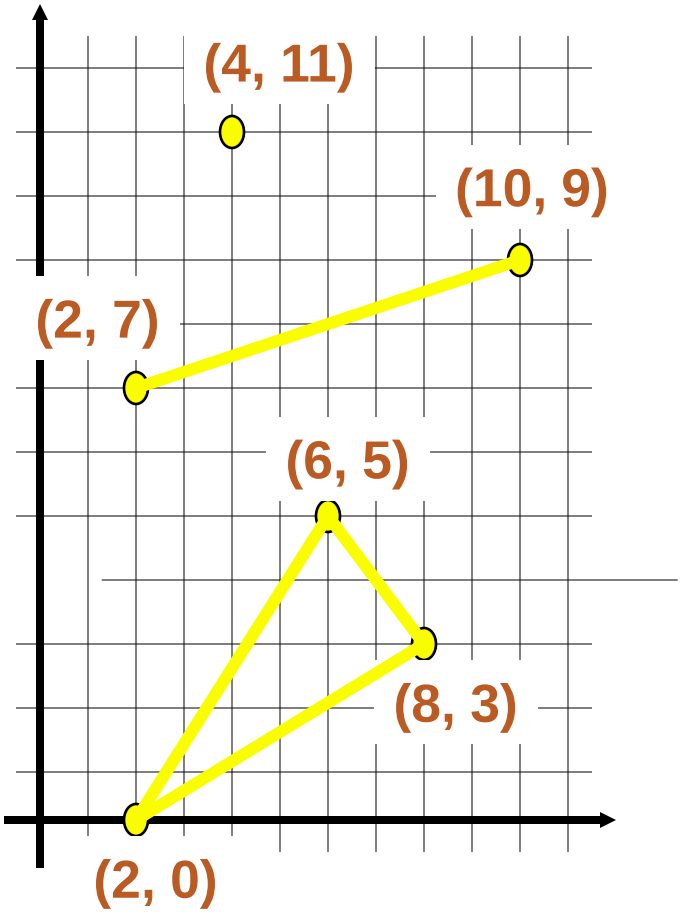
```
point P;
line L;
triangle T;
```



Nested structures



```
point P;
line L;
triangle T;

P.x = 4;

P.y = 11;
L.p1.x = 2;
L.p1.y = 7;
L.p2.x = 10;
L.p2.y = 9;
T.p1.x = 2;
T.p1.y = 0;
T.p2.x = 6;
T.p2.y = 5;
T.p3.x = 8;
T.p3.y = 3;
```



Structure as Functions

```
#include <iostream>
using namespace std;
struct Person
{
char name[50];
int age;
float salary;
};
void displayData(Person); // Function
declaration
int main()
{
Person p;
cout << "Enter Full name: ";
cin.get(p.name, 50);
cout << "Enter age: ";
cin >> p.age;
cout << "Enter salary: ";
cin >> p.salary;
// Function call with structure variable as an
argument
displayData(p);
return 0;
}
void displayData(Person p)
{
cout << "\nDisplaying Information." <<
endl;
cout << "Name: " << p.name << endl;
cout <<"Age: " << p.age << endl;
cout << "Salary: " << p.salary;
}
```

Structure as Functions

```
#include <iostream>
using namespace std;
struct Person {
char name[50];
int age;
float salary;
};
Person getData(Person);
void displayData(Person);
int main()
{
Person p;
p = getData(p);
displayData(p);
return 0;
}
Person getData(Person p) {
cout << "Enter Full name: ";
cin.get(p.name, 50);
cout << "Enter age: ";
cin >> p.age;
cout << "Enter salary: ";
cin >> p.salary;
return p;
}
void displayData(Person p)
{
cout << "\nDisplaying Information." <<
endl;
cout << "Name: " << p.name << endl;
cout <<"Age: " << p.age << endl;
cout << "Salary: " << p.salary;
}
```

Acknowledgements

Material in these slides has been taken from, the following resources

1. Deitel and Deitel: C++ How to Program, 7th Edition, Prentice Hall Publications
2. Robert Lafore: Object-Oriented Programming in C++, Fourth Edition, December 2001,Sams Publishing .
3. A Structured Programing Approach Using C++ by Behrouz A. Forouzan
4. www.cplusplus.com